

Character Literal Contd..

```
package com.ravi.character_literal;
```

```
//Java supports UNICODE
```

```
public class Test4
{
    void main()
    {
        String hindi = "\u0928\u092E\u0938\u094D\u0924\u0947";
        IO.println(hindi);

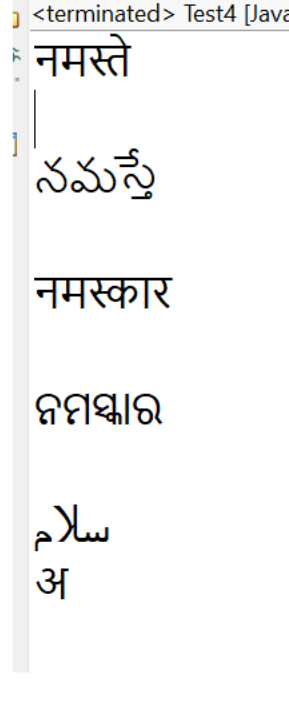
        String telugu = "\u0C28\u0C2E\u0C38\u0C4D\u0C24\u0C47";
        IO.println("\n"+telugu);

        String marathi = "\u0928\u092E\u0938\u094D\u0915\u093E\u0930";
        IO.println("\n"+marathi);

        String odia = "\u0B28\u0B2E\u0B38\u0B4D\u0B15\u0B3E\u0B30";
        IO.println("\n"+odia);

        String urdu = "\u0633\u0644\u0627\u0645";
        IO.println("\n"+urdu);

        IO.println("\u0905");
    }
}
```



```
Test5.java
-----
void main()
{
    int x = 'A';
    int y = 'B';

    IO.println(x+y); //131

    IO.println('A' + 'A'); // 130 [Numeric Promotion]

    IO.println("A"+"A"); //AA [String concatenation]
}
```

```
Test6.java
-----
//Range of UNICODE Value (0 - 65535) OR '\u0000' to '\uffff'
void main()
{
    char ch1 = 65535;
    IO.println("ch value is :"+ch1);

    char ch2 = 65536;
    IO.println("ch value is :"+ch2);
}
```

```
Test7.java
-----
//WAP in java to describe unicode representation of char in hexadecimal format

void main()
{
    int ch1 = '\u0000';
    IO.println(ch1);

    int ch2 = '\uffff';
    IO.println(ch2);

    char ch3 = '\u0041';
    IO.println(ch3); //A

    char ch4 = '\u0061';
    IO.println(ch4); //a
}
```

```
Test8.java
-----
void main()
{
    char c1 = 'A';
    char c2 = 65;
    char c3 = '\u0041';
    IO.println("c1 = "+c1+", c2 = "+c2+", c3 = "+c3);
}
```

```
Test9.java
-----
void main()
{
    int x = 'A';
    int y = '\u0041';
    IO.println("x = "+x+" y = "+y);
}
```

```
Test10.java
-----
//Every escape sequence is char literal

void main()
{
    char ch = '\n';
    IO.print("Hello");
    IO.println(ch);

    IO.println("Hello Java"); //Hello Java
    IO.println("\Hello learner!!!!\");
}
```

```
Test11.java
-----
void main()
{
    IO.println("Character size is :"+Character.SIZE);
    IO.println("Minimum value of Character :"+(int)Character.MIN_VALUE);
    IO.println("Maximum value of Character :"+(int)Character.MAX_VALUE);

    int c1 = '\u0000';
    int c2 = '\uffff';
    IO.println(c1+" : "+c2);
}
```

Boolean Literal :

- * It is used to represent two states i.e true OR false.
- * We have only one data type i.e boolean data type which can store **1 bit of information and the size of boolean data type depends upon JVM implementation**.
Example :

```
boolean isEmpty = true;
boolean isValid = false;
```
- * Unlike C & C++, we **cannot assign 0 and 1 to boolean data type because in java 0 & 1 represents integral literals**.
Example :

```
boolean isValid = 0; //Error [Valid in C & C++ but invalid in Java]
```

- * String value we cannot assign to boolean type.
Example :

```
boolean b = "true"; //error
```

```
Test1.java
-----
void main()
{
    boolean isValid = true;
    boolean isEmpty = false;

    IO.println(isValid);
    IO.println(isEmpty);
}
```

```
Test2.java
-----
void main()
{
    boolean c = 0;
    boolean d = 1;
    IO.println(c);
    IO.println(d);
}
```

```
Test3.java
-----
void main()
{
    boolean isValid = "true";
    IO.println(isValid);
}
```

String Literal :

String is a predefined class available in java.lang (Library) Package.

String is a collection of alpha-numeric character which is enclosed by double quotes. These characters can be alphabets, numbers, symbol or any special character.

Example:
"India"
"Hyderabad"
"B-61, S R Nagar"

In java we can create String object by using following 3 ways:

- 1) By using String Literal

```
String str1 = "india";
```
- 2) By using new keyword

```
String str2 = new String("Hyderabad");
```
- 3) By using Character array [Old Technique]

```
char ch[] = {'R', 'A', 'J'};
String s3 = new String(ch);
```

```
//Programs :
-----
Test1.java
-----
//String is collection of alpha-numeric character

void main()
{
    String x = "B-61 Hyderabad";
    IO.println(x);

    String y = "123";
    IO.println(y);

    String z = "67.90";
    IO.println(z);

    String p = "A";
    IO.println(p);
}
```

```
Test2.java
-----
//IQ
void main()
{
    String s = 15 + 29 + "Ravi" + 40 + 40;
    IO.println(s);
}
```

```
Test3.java
-----
//Three Ways to create the String Object

void main()
{
    //By using Literal
    String s1 = "Java";
    IO.println(s1);

    //By using new keyword
    String s2 = new String("Hyderabad");
    IO.println(s2);

    //Character Array
    char ch[] = {'R', 'A', 'J'};
    String s3 = new String(ch);
    IO.println(s3);
}
```

- 4) Punctuators:

It is also called separators.
- It is used to inform the compiler how things are grouped in the code.

() {} [] ; , . @